The authors present **CancerImmunoDB**, a manually curated database designed to catalog the immunomodulatory effects of anticancer therapies. Given the increasingly recognized role of the tumor microenvironment (TME) and immune modulation in determining therapeutic efficacy and patient prognosis, centralizing experimentally verified data is a highly relevant and timely endeavor. The abstract is well-structured and clearly defines the resource's purpose, scale, and features. The manuscript requires few major and minor changes.

Major Comments

1. **Missing Comparison with DIRMC Database:** The current work does not mention the DIRMC database (Database of Immunotherapy-Related Molecular Characteristics), which is a major recent resource in this field (Liu et al., 2024). The authors should include relevant references to DIRMC in the introduction or discussion sections. They also need to clearly explain how CancerImmunoDB is different—specifically highlighting that DIRMC focuses only on formal immunotherapies, whereas CancerImmunoDB uniquely focuses on the immune-altering side effects of *conventional* anticancer drugs.
2. **Access for Computer Models:** This database is a great fit for training computer models to predict drug reactions. However, the authors need to clearly state if researchers can download the entire dataset at once (like a CSV or JSON file) or use an API, rather than just searching for one drug at a time on the website.
3. **Standardizing Immune Terms:** The abstract mentions tracking altered "immune components," but the immune system is very complex. The authors should clarify if they use standard, widely accepted naming systems (like the Cell Ontology) so that all data is grouped consistently (e.g., making sure different types of T-cells or macrophages are labeled correctly).
4. **Exact Numbers vs. General Trends:** Please clarify how detailed the data is. Does the database only say if an immune cell type went "up" or "down" (qualitative), or does it give exact numbers like the specific drug dosage used or the exact amount of change (quantitative)?
5. **Ambiguity in Immune Modulation Categorization:** In the architecture figure the bottom-right box detailing the major fields, the database categorizes outcomes as *"Immune System Function Increased / Number Increased"* and *"Immune System Function Decreased / Number Decreased"*. However, Lumping "Function" and "Number" into a single category may obscure critical biological nuances. For example, a drug might induce proliferation of a specific immune cell (Number Increased) while simultaneously exhausting its cytotoxic capabilities (Function Decreased). The authors should clarify if users can query "Function" and "Number" independently, or if these are strictly coupled in the backend.

Furthermore, the category *"Negative immune regulations"* is ambiguous. The authors need to define if an increase in immunosuppressive cells (like Tregs or MDSCs) falls under this category, or if it is classified under general "Immune System Function Increased."

6. **Naming Consistency :**There is a clear inconsistency in the nomenclature of the database. The text of the abstract refers to the resource as **"CancerImmunoDB,"** whereas the central cylinder in the architecture diagram (and potentially the domain name) is labeled **"CanImmDB"**. The authors must ensure the name is standardized across the abstract, manuscript text, figures, and URL to avoid confusion.
7. **Keeping the Database Updated:** Since all the data is collected manually by reading papers, this process takes a lot of time. The authors need to explain what years of research are currently included and share their plan for how they will keep the database updated as new research is published.

Minor Comments

1. **Linking to Patient Data:** The database already connects to drug databases like PubChem and DrugBank, which is great. In the future, the authors might consider linking their drug data to patient gene expression databases (like TCGA) to make it even more useful.
2. **Acknowledging Bias:** In the discussion section, the authors should briefly mention that because they rely on published papers, their database might mostly show "positive" or "successful" results, since scientists rarely publish papers about drugs that had no effect.
3. **Missing Technical Stack Details:** While the architecture diagram excellently outlines the data sources, available tools, and query fields, it lacks technical transparency regarding the underlying software architecture. A standard bioinformatics database schematic should ideally indicate the **Data Tier** (e.g., MySQL, PostgreSQL, or MongoDB backend) and the **Application/Web Tier** (e.g., Django, Flask, PHP, Vue.js frontend). Adding a small layer to the diagram denoting the tech stack would greatly improve its technical rigor.
4. **Chemo-informatics Tool Clarification:** The "Modules" section highlights advanced computational tools, specifically *"3D Similarity Search"* and *"MACCS Key Based Search"*. Because 3D conformer generation from SMILES strings can vary wildly depending on the algorithm used, the authors should explicitly state in the manuscript (and perhaps briefly in the diagram's legend) which chemo-informatics libraries (e.g., RDKit, Open Babel, ChemAxon) power these backend structural calculations.

5. **Data Normalization and Units:** The "Major Fields" box lists **"Dose"** as a searchable parameter. Given that dosing in the literature is highly variable—ranging from μM or nM in *in vitro* cell lines to mg/kg in *in vivo* animal models—the authors should briefly explain how dosing units are normalized in the database schema to allow for accurate conditional searches, or if the dose is simply stored as a raw text string.